

INC-2026-0618 - Shipping Label Failover Did Not Activate

Incident Report | Engineering

Document ID	INC-2026-0618	Seller ID	S003
Department	Engineering	Classification	CONFIDENTIAL
Owner	Vikram Singh	Created Date	2026-06-20
Related Scenario ID	SC-005	Document Type	Incident Report

1. Incident Summary

Field	Value
Incident ID	INC-2026-0618
Severity	SEV-1
Service	Shipment Label Service
Component	Courier provider failover path
Detected	2026-06-18 06:50 IST
Mitigated	2026-06-18 08:20 IST
Resolved	2026-06-18 09:15 IST
Incident owner	Vikram Singh, Engineering
Operations owner	Neha Gupta
Seller affected	S003 - GreenCart Organics
Related ticket	TKT-2026-0618-233

The primary courier provider's label generation API degraded. Automatic failover to the backup provider did not activate because the stored backup credentials had expired, and the failover path attempted authentication only at the moment failover was required. The service fell back to retrying the degraded primary rather than reporting the credential failure. 2,100 orders were left without labels and 640 perishable orders missed their delivery slot.

2. Impact

Dimension	Detail
Orders in label-pending state	2,100
Orders that missed their delivery slot	640
Orders recovered into a later slot	1,460
Refunds issued	640, full value
Produce written off	approximately 640 subscription baskets
Duration of label generation outage	2 hours 40 minutes
Sellers affected	1 (S003)

The severity classification deserves explanation. 2,100 undispached orders would be SEV-2 on most accounts. On this account the goods are perishable and delivery is slot-based: a missed slot is a write-off, not a delay. The Seller Escalation Policy treats perishable fulfilment failures as SEV-1 by default and that default was applied here.

3. Detection

Detected by the seller, 70 minutes after label generation began failing.

GreenCart's dark store teams had picked and packed the morning subscription baskets and could not generate labels. The co-founder called Support at 06:50 IST, before the morning slot cut-off, and Support escalated at 07:12.

Platform monitoring showed the Shipment Label Service in a degraded state with a rising error rate from 05:40. An alert fired at 05:58 and was attributed to the known provider degradation, which the provider had itself announced on its status page. The alert was acknowledged and not escalated, because the expected behaviour in a provider degradation is automatic failover.

Nobody verified that failover had occurred. That is the detection failure in this incident: the alert was correctly raised, correctly attributed, and then closed on an assumption.

4. Timeline

Time (IST)	Event
18 Jun 05:40	SwiftLane API latency rises; error rate climbs
18 Jun 05:52	First label generation failures
18 Jun 05:58	Label service degradation alert fires; acknowledged as provider-side
18 Jun 06:10	Failover threshold breached; failover attempt begins
18 Jun 06:10	RapidPost authentication fails; failover aborts silently
18 Jun 06:11	Service resumes retrying SwiftLane
18 Jun 06:50	Seller calls Support before morning slot cut-off
18 Jun 07:12	Escalated to Engineering and Operations
18 Jun 07:26	Failover confirmed as not active despite threshold breach
18 Jun 07:35	Expired RapidPost credentials identified as the blocker
18 Jun 08:05	Credentials rotated with the backup provider
18 Jun 08:20	Failover executed manually; labels begin generating
18 Jun 09:15	Backlog of 2,100 labels cleared
18 Jun 10:00	640 orders confirmed as having missed their slot
18 Jun 14:00	Refunds issued for the 640 affected orders

5. Symptoms

- Label generation requests timing out after 30 seconds.
- Shipment Label Service reporting degraded, not failed.
- Retry counters climbing against the primary provider.
- No log entry indicating a failover attempt had been made or had failed.

- Orders accumulating in label-pending state with no terminal error.
- Provider status page confirming a known degradation at SwiftLane.

The absence of any failover log entry was the critical observation. The service had attempted failover at 06:10, but the failed authentication was caught by an exception handler that logged at debug level and returned control to the primary retry loop.

6. Affected Services

Service	Role	State
Shipment Label Service	Label generation and provider selection	Degraded; failover ineffective
SwiftLane integration (primary)	Label generation	Provider-side degradation
RapidPost integration (backup)	Failover label generation	Unavailable: expired credentials
Credential store	Held provider credentials	Functioning; expiry was visible but unmonitored
Order fulfilment pipeline	Consumed labels for dispatch	Blocked

7. Investigation

7.1 Why Failover Did Not Happen

The failover logic is threshold-based: when the primary provider's error rate exceeds 40 percent over a two-minute window, the service switches to the backup. That threshold was breached at 06:10 and the switch was attempted.

The backup provider requires an OAuth client credential exchange before label requests can be made. The stored client secret for RapidPost had an expiry date of 31 May 2026 and had not been renewed. The token exchange returned a 401.

The failover handler treated any exception during provider initialisation as a transient failure of that provider and reverted to the previously selected provider, which was the degraded primary. It logged the reversion at debug level.

So the service made a correct decision to fail over, discovered it could not, and then made the worst available choice silently.

7.2 Why The Expiry Was Not Noticed

The credential store recorded the expiry date. Nothing monitored it. There was no report, alert or dashboard covering credential expiry for provider integrations.

The backup provider had never been used in production. It was configured in October 2025 as open item OI-1 from the seller's onboarding project, and that item was closed on the evidence of a configuration entry rather than a successful failover test. In the eight months between configuration and this incident, no label had ever been generated through RapidPost in production.

An expired credential on a path that is never exercised is indistinguishable from a working configuration.

7.3 Why Falling Back Was Worse Than Failing

Had the service stopped and reported a terminal error at 06:10, the seller would have known within minutes rather than forty. The dark store teams could have held the pick, or the operations team could have arranged a manual dispatch for the highest-value baskets.

Instead the service continued retrying, producing exactly the signals a healthy-but-slow system produces. Everyone who looked at it, including the on-call engineer who acknowledged the 05:58 alert, saw a system responding to a known provider problem.

8. Root Cause

Automatic failover to the backup courier provider did not activate because the stored backup credentials had expired, and the failover path validated credentials only at the moment failover was attempted. The failover handler treated the resulting authentication failure as a transient provider error and silently reverted to the degraded primary provider, so label generation remained unavailable and no failure was surfaced.

9. Contributing Factors

1. **Failover path never exercised in production** in eight months of configuration.
2. **No credential expiry monitoring** for provider integrations.
3. **Silent reversion on failover failure**, logged at debug level.
4. **Onboarding item closed on configuration evidence** rather than on a successful test.
5. **Alert acknowledged on an assumption** that failover would handle a known provider degradation.
6. **No end-to-end label generation health check** independent of provider selection.

10. Mitigation

Credentials were rotated with the backup provider at 08:05 and failover was executed manually at 08:20. Labels began generating immediately and the backlog of 2,100 cleared by 09:15.

There was no faster mitigation available. The primary provider remained degraded until approximately 11:00, so the backup was the only viable path, and the backup could not be used until its credentials were valid.

Time from escalation to mitigation: 68 minutes, of which 23 minutes were spent identifying that failover had not occurred and 30 minutes obtaining rotated credentials from the provider.

11. Resolution

Action	Completed
Backup provider credentials rotated	2026-06-18 08:05
Manual failover executed	2026-06-18 08:20
Label backlog cleared	2026-06-18 09:15
640 refunds issued under the refund-on-report exception	2026-06-18 14:00
Credential expiry monitoring for all provider integrations	2026-06-26
Failover pre-flight check on a scheduled basis	2026-07-03
Failover failure raised as a terminal error, not a debug log	2026-07-03

Action	Completed
Live failover test performed and passed	2026-07-15

12. Validation

Check	Method	Result
Failover works end to end	Live test, primary disabled deliberately, 15 July	Failover in 94 seconds; labels generated through backup
Expiry monitoring effective	Test credential set to expire in 7 days	Alert raised at 30, 14 and 7 days
Pre-flight check effective	Backup credential deliberately invalidated	Alert raised within one scheduled cycle, before any failure
Failover failure now visible	Backup made unreachable during a forced failover	Terminal error raised; no silent reversion
Label generation health	30-day observation	No further label-pending accumulation

The live failover test on 15 July also satisfied milestone M3 in the seller's conditional tier upgrade decision, which requires a verified failover with a live test rather than a configuration check.

13. Preventive Actions

ID	Action	Owner	Target	Status
PA-1	Credential expiry monitoring across provider integrations	Neha Gupta	2026-06-26	Complete
PA-2	Scheduled failover pre-flight validation	Vikram Singh	2026-07-03	Complete
PA-3	Failover failure treated as terminal and alerted	Vikram Singh	2026-07-03	Complete
PA-4	Quarterly live failover test for fulfilment-critical paths	Neha Gupta	2026-07-15	Complete
PA-5	Onboarding runbook: failover items require a live test to close	Neha Gupta	2026-07-24	Complete
PA-6	End-to-end label generation synthetic check every 5 minutes	Vikram Singh	2026-08-07	Complete

14. Follow-Up

The broader finding is that a redundancy that is never exercised is not redundancy. PA-4 and PA-5 address this directly for fulfilment paths, and the Seller Onboarding Runbook now refuses to close a failover item on configuration evidence alone.

Post-incident review held 20 June 2026 with Engineering, Operations and Support. The review noted that the platform had told this seller it had a failover, and that the seller had reasonably relied on that statement for eight months.